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Fermatean Triple-Valued Neutrosophic Soft Sets and Their Associated Soft Topological Spaces

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Abstract: Parameterized uncertainty models need to represent support, opposition, and unresolved evidence while retaining soft-set parameterization. This paper develops a Fermatean Triple-Valued Neutrosophic Soft Set (FTVNSS) in which each object-parameter assessment has five coordinates: truth, relative truth, hesitation, relative falsity, and falsity. Cubic Fermatean restrictions apply separately to the truth–falsity and relative-truth–relative-falsity pairs, while hesitation remains an independent unit-interval degree. Inclusion, complement, union, intersection, and AND/OR products are formulated through coordinatewise lattice operations. Numerical examples verify the stated calculations, and a supremum/infimum argument establishes admissibility for arbitrary unions and intersections. A corresponding Fermatean Triple-Valued Neutrosophic Soft Topological Space (FTVNSTS) is introduced, with ordinary interior and closure defined before semi-open, pre-open, and b-open sets. Pre-interior, pre-closure, exterior, frontier, and p-closed mappings are then developed. The finite-intersection condition on pre-open sets is stated explicitly for the identities that require it, and the frontier inclusions are proved in detail. The framework combines soft parameterization, an explicit hesitation coordinate, and a topological layer; its contribution is theoretical rather than an empirically validated ranking procedure.

Keywords: Fermatean fuzzy set; Neutrosophic soft set; Triple-valued neutrosophic soft set; Soft topology; Pre-open set; Uncertainty; Operators

1 Introduction

Uncertain information is rarely exhausted by a single degree of membership. Fuzzy sets provide graded membership [1], while intuitionistic fuzzy sets separate support from non-support [2]. Neutrosophic formulations introduce an independent indeterminacy component [3]; single-valued neutrosophic sets constrain these assessments to the unit interval for practical use [4]. Soft set theory contributes a different but complementary idea by organizing uncertainty through parameters rather than through a single global membership function [5]. Neutrosophic soft sets [6] and double-valued neutrosophic models [7] demonstrate how these perspectives could be combined. Fermatean fuzzy sets further enlarge the admissible region for paired positive and negative assessments through a cubic condition [8].

Recent work has expanded the Fermatean framework in several directions. Fermatean fuzzy soft aggregation operators have been developed for parameterized decision information [9], while Fermatean fuzzy soft expert sets incorporate multiple evaluators [10]. Complex Fermatean fuzzy N-soft sets use a finite ordinal rating scale with N levels, together with complex-valued grades that encode magnitude and phase [11]. Fermatean fuzzy bipolar soft sets represent positive and negative parameter spaces jointly [12]. These studies combine Fermatean admissibility with richer information structures and parameter mechanisms.

Two surveys describe this broadening of scope. Büyüközkan et al. reviewed Fermatean fuzzy extensions and identified continuing growth in hybrid representations and decision models [13], while Alcantud et al. [14] surveyed soft-set theory and highlighted the sustained development of soft topology and algebraic structures. At the applied level, similarity and entropy measures [15], Fermatean fuzzy ranking methods [16], healthcare waste selection

models [17], and Fermatean fuzzy soft decision support for solar projects [18] illustrate the demand for uncertainty representations that preserve more information before aggregation. Meanwhile, Aydin demonstrates how new Fermatean fuzzy theories could be embedded in multi-criteria decision procedures [19].

The neutrosophic-soft literature has developed along a parallel trajectory. Saeed et al. [20] formulated a Fermatean neutrosophic soft model with algebraic and topological components, and Alkhazaleh et al. [21] introduced a Fermatean neutrosophic soft set with decision-making operations. Most directly related to the present work, Hatamleh et al. [22] developed Fermatean Double-Valued Neutrosophic Soft Sets (FDVNSSs) and an associated soft-topological framework. Soft-set parameterization also supports topological structures defined through arbitrary unions and finite intersections [23].

The double-valued predecessor [22] distinguishes absolute and relative truth/falsity information, but it does not retain a separate hesitation coordinate inside the same soft record. The proposed Fermatean Triple-Valued Neutrosophic Soft Set (FTVNSS) adds that coordinate and develops the corresponding algebraic and topological constructions. The contribution is deliberately theoretical: the objective is to establish a coherent structure that could later support ranking, aggregation, diagnosis, and group-decision algorithms.

The main contributions are summarized as follows:

1. A five-coordinate FTVNSS representation is defined using two Fermatean truth–falsity pair constraints, with an explicit hesitation component kept separate from both pairs.
2. A coordinate-wise order is specified, and complement, union, intersection, and AND/OR products are shown to preserve admissibility and satisfy the expected lattice identities.
3. An FTVNSS soft topology is constructed, together with generalized open sets and pre-interior, pre-closure, exterior, and frontier properties; the assumptions required for finite-intersection identities are stated explicitly.

2 Literature Review and Mathematical Background

The literature can be organized into three streams relevant to this study. The first develops Fermatean fuzzy soft and expert structures. The aggregation framework combines Fermatean fuzzy numbers through soft parameters [9], whereas the expert model incorporates multiple evaluators [10]. The N-soft extension uses ordinal parameter grades [11], and the bipolar soft construction represents paired positive and negative parameter descriptions [12]. These models motivate a parameterized Fermatean environment, but do not provide the five-coordinate topology developed here.

The second stream concerns measurement and decision procedures. Similarity and entropy functions have been proposed for Fermatean fuzzy information [15], while multi-criteria decision-making (MCDM) studies apply Fermatean representations to ranking, healthcare technology selection, and energy planning [16–19]. These studies indicate how an uncertainty representation can support scoring, distance, aggregation, or optimization. The present paper concentrates on the set structure and topology that precede such algorithms.

The third stream combined neutrosophic, soft, and topological ideas. Fermatean neutrosophic soft sets have recently been studied from algebraic and decision-making perspectives [20, 21]. The closest precursor was the Fermatean double-valued neutrosophic soft topology [22], where generalized open sets and derived topological notions were developed. The proposed model differed by retaining an additional hesitation coordinate throughout the soft-set record. This distinction is important when unresolved evidence should not be absorbed into either a truth-type or falsity-type assessment.

For notation, let X be a nonempty universe and E be a nonempty parameter set. A soft set was interpreted as a parameter-indexed family of information records. We adopted the standard soft-set viewpoint of Molodtsov [5], the neutrosophic-soft viewpoint of Maji [6], and the intuitionistic fuzzy soft formulation of Maji et al. [24] only as background. Rather than restating those definitions, we formulated the proposed construction directly through the five-coordinate record assigned to each pair (e, x) in $E \times X$.

The proposed construction must establish its information record and topological operators before any scoring or aggregation stage is introduced. Soft topological spaces [23] provide a parameterized topological framework, while Fermatean neutrosophic topological spaces [25] combine Fermatean uncertainty records with topological axioms. Their assumptions must be examined against the five-coordinate order and complement adopted here. Accordingly, the present paper verifies admissibility of its operations directly and states separately the additional condition used for finite-intersection results on pre-open sets. No decision rule is claimed or evaluated.

3 Fermatean Double-Valued Neutrosophic Soft Sets as a Predecessor

The FDVNSS and its associated Fermatean Double-Valued Neutrosophic Soft Topological Space (FDVNSTS) [22] are the immediate structural predecessors of the present construction. The model uses four coordinates for truth, relative truth, relative falsity, and falsity, together with soft parameters and Fermatean restrictions. The present study retains that parameterized viewpoint and introduces a separate hesitation coordinate. To ensure closure under the max/min operations below, the truth–falsity and relative-truth–relative-falsity channels are treated as Fermatean

pairs [8], while hesitation is stored independently. Table 1 summarizes the resulting structural differences between the proposed model and representative existing frameworks.

Table 1. Position of the proposed model relative to recent representative frameworks

Framework	Information Record	Soft/Parameter Layer	Representative Emphasis
Fermatean fuzzy set [8]	Membership/non-membership	No	Cubic admissibility for paired grades
Fermatean fuzzy soft/expert models [9, 10]	Fermatean grades, optionally expert-indexed	Yes	Aggregation and group decision information
Complex/N-soft variants [11]	Fermatean information with richer parameter coding	Yes	Ordered or multi-dimensional parameter information
Fermatean neutrosophic soft sets [20, 21]	Neutrosophic coordinates under Fermatean constraints	Yes	Hybrid uncertainty and decision operations
FDVNSS/FDVNSTS [22]	Four truth/falsity-side coordinates	Yes + topology	Double-valued neutrosophic soft topology
Proposed FTVNSS/FTVNSTS	Five coordinates including explicit hesitation	Yes + topology	Five-coordinate soft topology and derived operators

Note: FDVNSS = Fermatean Double-Valued Neutrosophic Soft Set; FDVNSTS = Fermatean Double-Valued Neutrosophic Soft Topological Space; FTVNSS = Fermatean Triple-Valued Neutrosophic Soft Set; FTVNSTS = Fermatean Triple-Valued Neutrosophic Soft Topological Space. The table compares the information record, the presence of a soft/parameter layer, and the representative emphasis of each framework.

4 Fermatean Triple-Valued Neutrosophic Soft Sets

This section defines the five-coordinate structure. A record is written as $\langle T, RT, H, RF, F\rangle$, where T is truth, RT is relative truth, H is hesitation, RF is relative falsity, and F is falsity. Each operation is applied pointwise to every parameter-object pair.

Definition 4.1 (FTVNSS) Let X and E be nonempty sets. A FTVNSS (F, E) assigns to each $(e, x) \in E \times X$ a five-coordinate record:

$$v_F(e, x) = \langle T_F(e, x), R_{T,F}(e, x), H_F(e, x), R_{F,F}(e, x), F_F(e, x)\rangle \in [0, 1]^5.$$

A record is admissible when the two Fermatean pair conditions below hold:

$$T^3 + F^3 \le 1, \qquad R_T^3 + R_F^3 \le 1, \qquad 0 \le H \le 1.$$

Both the absolute truth–falsity pair and the relative truth–falsity pair satisfy a Fermatean cubic restriction, while hesitation is treated as an independent coordinate. In particular, H is not reconstructed from the other four entries.

Example 4.1 Let $X = \{x_1, x_2\}$ describe two alternatives and $E = \{e_1, e_2\}$ two evaluation parameters. Consider the following admissible records:

$$\begin{aligned} F(e_1): &\quad x_1 \mapsto \langle 0.44, 0.18, 0.21, 0.12, 0.36\rangle, \quad x_2 \mapsto \langle 0.31, 0.24, 0.16, 0.20, 0.48\rangle; \\ F(e_2): &\quad x_1 \mapsto \langle 0.58, 0.14, 0.19, 0.10, 0.27\rangle, \quad x_2 \mapsto \langle 0.26, 0.29, 0.23, 0.17, 0.52\rangle. \end{aligned}$$

For the first record, $0.44^3 + 0.36^3 = 0.131840 < 1$ and $0.18^3 + 0.12^3 = 0.007560 < 1$, while $H = 0.21$. Thus both Fermatean pair constraints are satisfied.

Remark 4.2 (Order and admissibility) The pairwise Fermatean restrictions are preserved by the complement and max/min operations below. The coordinate order increases in T and RT and decreases in H, RF, and F.

Definition 4.3 (Inclusion) For admissible records a and b, write $a \preccurlyeq b$ when $T_a \le T_b$, $R_{T,a} \le R_{T,b}$, $H_a \ge H_b$, $R_{F,a} \ge R_{F,b}$, and $F_a \ge F_b$. For FTVNSSs (F, E) and (G, E), write $(F, E) \subseteq (G, E)$ when $v_F(e, x) \preccurlyeq v_G(e, x)$ for every $e \in E$ and $x \in X$.

Example 4.4 Let $a = \langle 0.44, 0.18, 0.21, 0.12, 0.36\rangle$ and $d = \langle 0.53, 0.22, 0.18, 0.10, 0.30\rangle$. Both are admissible and $a \preccurlyeq d$ because the two positive coordinates increase while H, R_F, and F decrease. Applying the same comparison to every object-parameter pair yields soft-set inclusion.

Definition 4.5 (Null FTVNSS) The null FTVNSS, denoted 0_E, assigns the least record $\langle 0, 0, 1, 1, 1\rangle$ to every object under every parameter:

$$0_E = \{(e, \{\langle x, 0, 0, 1, 1, 1\rangle : x \in X\}) : e \in E\}.$$

Example 4.6 For any finite X and E, assigning $\langle 0,0,1,1,1\rangle$ to every object-parameter pair produces 0_E. Under the order in Definition 4.3, 0_E is contained in every FTVNSS.

Definition 4.7 (Absolute FTVNSS) The absolute FTVNSS, denoted 1_E, assigns the greatest record $\langle 1,1,0,0,0\rangle$ to every object under every parameter:

$$1_E = \{(e, \{\langle x,1,1,0,0,0\rangle : x \in X\}) : e \in E\}.$$

Example 4.8 If each object-parameter pair is assigned $\langle 1,1,0,0,0\rangle$, the resulting soft set is 1_E. Every admissible FTVNSS is contained in 1_E.

Definition 4.9 (Complement) For a record $a = \langle T, R_T, H, R_F, F\rangle$, define its complement by reversing the truth/falsity sides and complementing hesitation:

$$a^{\mathrm{c}} = \langle F, R_F, 1-H, R_T, T\rangle.$$

Example 4.10 For the first record of Example 4.1, $\langle 0.44, 0.18, 0.21, 0.12, 0.36\rangle^{\mathrm{c}} = \langle 0.36, 0.12, 0.79, 0.18, 0.44\rangle$. Applying the same rule a second time returns the original record.

Definition 4.11 (Union) Let $a = \langle T_1, R_{T1}, H_1, R_{F1}, F_1\rangle$ and $b = \langle T_2, R_{T2}, H_2, R_{F2}, F_2\rangle$. Their union record is:

$$a \cup b = \langle \max(T_1,T_2), \max(R_{T1},R_{T2}), \min(H_1,H_2), \min(R_{F1},R_{F2}), \min(F_1,F_2)\rangle.$$

Example 4.12 With $a = \langle 0.44, 0.18, 0.21, 0.12, 0.36\rangle$ and $b = \langle 0.53, 0.11, 0.26, 0.16, 0.29\rangle$, the union is:

$$a \cup b = \langle 0.53, 0.18, 0.21, 0.12, 0.29\rangle.$$

Definition 4.13 (Intersection) For the same records, the intersection is defined by minima on the two positive coordinates and maxima on hesitation and the two falsity-side coordinates:

$$a \cap b = \langle \min(T_1,T_2), \min(R_{T1},R_{T2}), \max(H_1,H_2), \max(R_{F1},R_{F2}), \max(F_1,F_2)\rangle.$$

Example 4.14 For the two records used above,

$$a \cap b = \langle 0.44, 0.11, 0.26, 0.16, 0.36\rangle.$$

Proposition 4.15 For FTVNSSs (F,E), (G,E), and (H,E), complement, union, and intersection preserve admissibility. Moreover, union and intersection are commutative and associative, with 0_E and 1_E as their respective neutral elements:

$$F \cup 0_E = F, \quad F \cap 1_E = F, \quad F \cup G = G \cup F, \quad F \cap G = G \cap F,$$
$$F \cup (G \cup H) = (F \cup G) \cup H, \quad F \cap (G \cap H) = (F \cap G) \cap H.$$

Proof Under complementation, each Frematean pair is reversed and H is replaced by $1-H$; therefore, admissibility is preserved. For a union, if $\max(T_a,T_b) = T_a$, then $\min(F_a,F_b) \le F_a$ and hence $\max(T_a,T_b)^3 + \min(F_a,F_b)^3 \le T_a^3 + F_a^3 \le 1$; the same argument applies to the relative pair. The intersection argument is dual: if $\max(F_a,F_b) = F_a$, then $\min(T_a,T_b) \le T_a$. The H-coordinate remains in $[0,1]$ under both min and max. The neutral, commutative, and associative identities then follow coordinatewise.

Proposition 4.16 (De Morgan laws) For FTVNSSs (F,E) and (G,E),

$$(F \cap G)^{\mathrm{c}} = F^{\mathrm{c}} \cup G^{\mathrm{c}}, \qquad (F \cup G)^{\mathrm{c}} = F^{\mathrm{c}} \cap G^{\mathrm{c}}.$$

Proof The complement exchanges the truth/falsity sides, swaps the corresponding relative coordinates, and replaces H by $1-H$. The identities therefore reduce to the scalar relations $1-\min(u,v) = \max(1-u,1-v)$ and $1-\max(u,v) = \min(1-u,1-v)$, together with the max/min definitions on the remaining coordinates.

Definition 4.17 (AND product) If (F,A) and (G,B) are FTVNSSs on the same universe X, define $(F,A) \wedge (G,B) = (H, A\times B)$, where $v_H((a,b),x) = v_F(a,x) \cap v_G(b,x)$ for every $(a,b,x) \in A \times B \times X$.

Example 4.18 At a fixed object x, suppose $v_F(a_1,x) = a$ and $v_G(b_1,x) = b$ are the records of Example 4.12. Then $v_H((a_1,b_1),x) = a \cap b = \langle 0.44, 0.11, 0.26, 0.16, 0.36\rangle$. The same rule applies to all other parameter-object combinations.

Definition 4.19 (OR product) If (F,A) and (G,B) are FTVNSSs on X, define $(F,A) \vee (G,B) = (K, A\times B)$, where $v_K((a,b),x) = v_F(a,x) \cup v_G(b,x)$ for every $(a,b,x) \in A \times B \times X$.

Example 4.20 For the records and object used in Example 4.18,

$$v_K((a_1,b_1),x) = a \cup b = \langle 0.53, 0.18, 0.21, 0.12, 0.29\rangle.$$

The AND and OR products therefore use the pointwise intersection and union rules without an additional numerical operation.

5 Fermatean Triple-Valued Neutrosophic Soft Topological Spaces

The admissible FTVNSSs form a complete lattice whose finite union and intersection operations are distributive. For a nonempty family $\{A_i : i \in I\}$, arbitrary unions and intersections are defined pointwise by taking suprema on the increasing coordinates and infima on the decreasing coordinates for union, and the reverse for intersection. Writing U for the arbitrary union and V for the arbitrary intersection, their records at (e, x) are:

$$v_U(e,x) = \langle \sup_i T_i,\ \sup_i R_{T,i},\ \inf_i H_i,\ \inf_i R_{F,i},\ \inf_i F_i \rangle,$$
$$v_V(e,x) = \langle \inf_i T_i,\ \inf_i R_{T,i},\ \sup_i H_i,\ \sup_i R_{F,i},\ \sup_i F_i \rangle.$$

The empty union is 0_E and the empty intersection is 1_E. These operations preserve admissibility. For unions, put $f^* = \inf_i F_i$. Then $T_i^3 + f^{*3} \le T_i^3 + F_i^3 \le 1$ for every i. Taking the supremum gives $(\sup_i T_i)^3 + f^{*3} \le 1$. The same argument applies to the relative pair. For intersections, put $t^* = \inf_i T_i$ and take the supremum in $t^{*3} + F_i^3 \le 1$. Hesitation remains in $[0, 1]$. All identities are pointwise, so the topological axioms below are well defined. Finite distributivity follows from the scalar min/max identities, and arbitrary De Morgan laws follow from the supremum/infimum definitions.

Definition 5.1 (FTVNSS topology) Let $\mathrm{FTVNSS}(X, E)$ denote the family of all FTVNSSs on (X, E), and let $\tau \subseteq \mathrm{FTVNSS}(X, E)$. We called τ an FTVNSS topology when three conditions held: (i) 0_E and 1_E belong to τ; (ii) the union of any subfamily of τ belongs to τ; and (iii) the intersection of every finite subfamily of τ belongs to τ. A triple (X, τ, E) satisfying these conditions was called a Fermatean Triple-Valued Neutrosophic Soft Topological Space (FTVNSTS).

Example 5.2 Let F and G be FTVNSSs such that $F \cap G = 0_E$ and $F \cup G = 1_E$. Then, $\tau = \{0_E, 1_E, F, G\}$ is closed under the unions and finite intersections generated by its members and is therefore an FTVNSS topology.

Theorem 5.3 If τ_1 and τ_2 are FTVNSS topologies on (X, E), then $\tau_1 \cap \tau_2$ is also an FTVNSS topology.

Proof Both 0_E and 1_E belong to each topology and therefore to their intersection. Any union of sets belonging to both τ_1 and τ_2 remains in each topology by the union axiom. The same reasoning applies to finite intersections. Hence $\tau_1 \cap \tau_2$ satisfies all three topology axioms.

Remark 5.4 The set-theoretic union $\tau_1 \cup \tau_2$ need not be a topology: a finite intersection formed from one member of τ_1 and one member of τ_2 may fail to lie in either family.

For an FTVNSS A, the ordinary interior and closure in (X, τ, E) are:

$$\mathrm{Int}(A) = \bigcup \{O \in \tau : O \subseteq A\}, \qquad \mathrm{Cl}(A) = \bigcap \{C : A \subseteq C \text{ and } C^{\mathrm{c}} \in \tau\}.$$

where, $\bigcup$ and $\bigcap$ denote the arbitrary-family operations defined above. Thus $\mathrm{Int}(A)$ is the largest open FTVNSS contained in A, and $\mathrm{Cl}(A)$ is the smallest closed FTVNSS containing A. Both operators are monotone. By complementation, $[\mathrm{Cl}(A)]^{\mathrm{c}} = \mathrm{Int}(A^{\mathrm{c}})$ and $[\mathrm{Int}(A)]^{\mathrm{c}} = \mathrm{Cl}(A^{\mathrm{c}})$.

Definition 5.5 (Generalized openness) Let (X, τ, E) be an FTVNSTS and let A be an FTVNSS. We call A semi-open when $A \subseteq \mathrm{Cl}(\mathrm{Int}(A))$; pre-open (p-open) when $A \subseteq \mathrm{Int}(\mathrm{Cl}(A))$; and b-open when $A \subseteq \mathrm{Cl}(\mathrm{Int}(A)) \cup \mathrm{Int}(\mathrm{Cl}(A))$. A set is p-closed when its complement is p-open; the other closed variants are defined dually.

Arbitrary unions of p-open FTVNSSs are p-open. If A is the union of a family $\{A_i : i \in I\}$ of p-open FTVNSSs, monotonicity gives $A_i \subseteq \mathrm{Int}(\mathrm{Cl}(A_i)) \subseteq \mathrm{Int}(\mathrm{Cl}(A))$ for every i. Hence $A \subseteq \mathrm{Int}(\mathrm{Cl}(A))$. The empty union is p-open because $\mathrm{Int}(\mathrm{Cl}(0_E)) = 0_E$. Arbitrary De Morgan laws then show that arbitrary intersections of p-closed FTVNSSs are p-closed. Consequently, the largest and smallest sets in Definition 5.6 exist.

Definition 5.6 (p-interior and p-closure) The p-interior $\mathrm{pInt}(A)$ is the union of all p-open FTVNSSs contained in A; equivalently, it is the largest p-open subset of A. The p-closure $\mathrm{pCl}(A)$ is the intersection of all p-closed FTVNSSs containing A; equivalently, it is the smallest p-closed superset of A.

Definition 5.7 (p-frontier) The p-frontier of A is $\mathrm{Fr}(A) = \mathrm{pCl}(A) \cap \mathrm{pCl}(A^{\mathrm{c}})$.

Definition 5.8 (Exterior) The exterior of A is $\mathrm{Ext}(A) = \mathrm{pInt}(A^{\mathrm{c}})$.

Definition 5.9 (p-closed mapping) Let (X, τ, E) and (Y, σ, K) be FTVNSTSs, with maps $f : X \to Y$ and $\varphi : E \to K$. For an FTVNSS A on (X, E), define its image $(f, \varphi)_* A$ on (Y, K) as follows. At each (k, y), the image record is the admissible coordinatewise union of the source records $v_A(e, x)$ over all pairs satisfying $\varphi(e) = k$ and $f(x) = y$. An empty fibre receives the null record $\langle 0, 0, 1, 1, 1 \rangle$. The mapping (f, φ) is p-closed when this image is p-closed in (Y, σ, K) for every p-closed A in (X, τ, E).

Theorem 5.10 An FTVNSS A is p-open if and only if $\mathrm{pInt}(A) = A$.

Proof If A is p-open, then A is one of the sets whose union defines $\mathrm{pInt}(A)$, so $A \subseteq \mathrm{pInt}(A)$; the reverse inclusion follows from the definition of p-interior. Conversely, if $\mathrm{pInt}(A) = A$, then A is p-open because $\mathrm{pInt}(A)$ is p-open.

Standing assumption for the finite-operation identities. The family of p-open FTVNSSs is assumed to be closed under finite intersections; equivalently, p-closed FTVNSSs are closed under finite unions. This condition

is needed for Theorem 5.11(iv), Theorem 5.13(iv), and statements (1), (3), and (4) of Theorem 5.15. The other displayed properties, Theorems 5.10 and 5.12, and the complement duality of Theorem 5.14 hold without this additional condition.

The assumption is not automatic. To see this, let E be a singleton and represent ordinary subsets of $X = \{a, b, c\}$ by absolute records at their members and null records elsewhere. In the topology $\tau = \{0_E, \{a, b\}, 1_E\}$, the sets $\{a, c\}$ and $\{b, c\}$ are p-open, but their intersection $\{c\}$ is not p-open. Here the subset notation denotes its corresponding FTVNSS embedding.

Theorem 5.11 (p-interior properties) For FTVNSSs A and B, the following relations hold under the standing assumption:

$$\begin{gathered}\text{(i) } \mathrm{pInt}(\mathrm{pInt}(A)) = \mathrm{pInt}(A);\\ \text{(ii) } \mathrm{pInt}(0_E) = 0_E \text{ and } \mathrm{pInt}(1_E) = 1_E;\\ \text{(iii) } A \subseteq B \Rightarrow \mathrm{pInt}(A) \subseteq \mathrm{pInt}(B);\\ \text{(iv) } \mathrm{pInt}(A \cap B) = \mathrm{pInt}(A) \cap \mathrm{pInt}(B);\\ \text{(v) } \mathrm{pInt}(A) \cup \mathrm{pInt}(B) \subseteq \mathrm{pInt}(A \cup B).\end{gathered}$$

Proof Idempotence and the null/absolute identities follow from the characterization of $\mathrm{pInt}(A)$ as the largest p-open subset of A. Monotonicity follows because every p-open subset of A is also a subset of B whenever $A \subseteq B$. Under the standing assumption, $\mathrm{pInt}(A) \cap \mathrm{pInt}(B)$ is p-open and is the largest p-open subset contained in both A and B, which proves (iv). Finally, $\mathrm{pInt}(A)$ and $\mathrm{pInt}(B)$ are p-open subsets of $A \cup B$, yielding (v).

Theorem 5.12 An FTVNSS A is p-closed if and only if $\mathrm{pCl}(A) = A$.

Proof If A is p-closed, it is already a p-closed superset of itself and minimality of $\mathrm{pCl}(A)$ gives equality. The converse follows immediately because $\mathrm{pCl}(A)$ is p-closed.

Theorem 5.13 (p-closure properties) For FTVNSSs A and B, the following relations hold; statement (iv) uses the standing assumption:

$$\begin{gathered}\text{(i) } \mathrm{pCl}(\mathrm{pCl}(A)) = \mathrm{pCl}(A);\\ \text{(ii) } \mathrm{pCl}(0_E) = 0_E \text{ and } \mathrm{pCl}(1_E) = 1_E;\\ \text{(iii) } A \subseteq B \Rightarrow \mathrm{pCl}(A) \subseteq \mathrm{pCl}(B);\\ \text{(iv) } \mathrm{pCl}(A \cup B) = \mathrm{pCl}(A) \cup \mathrm{pCl}(B);\\ \text{(v) } \mathrm{pCl}(A \cap B) \subseteq \mathrm{pCl}(A) \cap \mathrm{pCl}(B).\end{gathered}$$

Proof The statements are dual to Theorem 5.11. In particular, $\mathrm{pCl}(A)$ is the smallest p-closed superset of A, and finite unions of p-closed sets remain p-closed under the standing assumption.

Theorem 5.14 (Complement duality) For every FTVNSS A,

$$[\mathrm{pCl}(A)]^{\mathrm{c}} = \mathrm{pInt}(A^{\mathrm{c}}), \qquad [\mathrm{pInt}(A)]^{\mathrm{c}} = \mathrm{pCl}(A^{\mathrm{c}}).$$

Proof Complementation reverses inclusion and exchanges p-open and p-closed sets. Taking complements of the smallest p-closed superset therefore yields the largest p-open subset of the complement, and vice versa.

Theorem 5.15 (Exterior and frontier) Let A and B be FTVNSSs in an FTVNSTS satisfying the standing assumption. Then,

$$\begin{gathered}\text{(i) } \mathrm{Ext}(A \cup B) = \mathrm{Ext}(A) \cap \mathrm{Ext}(B);\\ \text{(ii) } \mathrm{Ext}(A \cap B) \supseteq \mathrm{Ext}(A) \cup \mathrm{Ext}(B);\\ \text{(iii) } \mathrm{Fr}(A \cup B) \subseteq \mathrm{Fr}(A) \cup \mathrm{Fr}(B);\\ \text{(iv) } \mathrm{Fr}(A \cap B) \subseteq \mathrm{Fr}(A) \cup \mathrm{Fr}(B).\end{gathered}$$

Proof De Morgan's law and Theorem 5.11(iv) give:

$$\mathrm{Ext}(A \cup B) = \mathrm{pInt}(A^{\mathrm{c}} \cap B^{\mathrm{c}}) = \mathrm{Ext}(A) \cap \mathrm{Ext}(B),$$

which proves (i). Monotonicity gives:

$$\mathrm{Ext}(A \cap B) = \mathrm{pInt}(A^{\mathrm{c}} \cup B^{\mathrm{c}}) \supseteq \mathrm{Ext}(A) \cup \mathrm{Ext}(B),$$

proving (ii). For (iii), the finite-union identity for pCl, monotonicity, and finite distributivity yield:

$$\begin{aligned}\mathrm{Fr}(A \cup B) &= [\mathrm{pCl}(A) \cup \mathrm{pCl}(B)] \cap \mathrm{pCl}(A^{\mathrm{c}} \cap B^{\mathrm{c}})\\ &\subseteq [\mathrm{pCl}(A) \cup \mathrm{pCl}(B)] \cap \mathrm{pCl}(A^{\mathrm{c}}) \cap \mathrm{pCl}(B^{\mathrm{c}})\\ &= [\mathrm{Fr}(A) \cap \mathrm{pCl}(B^{\mathrm{c}})] \cup [\mathrm{Fr}(B) \cap \mathrm{pCl}(A^{\mathrm{c}})]\\ &\subseteq \mathrm{Fr}(A) \cup \mathrm{Fr}(B).\end{aligned}$$

For (iv), the same properties give:

$$\begin{aligned}\mathrm{Fr}(A \cap B) &= \mathrm{pCl}(A \cap B) \cap \mathrm{pCl}(A^{\mathrm{c}} \cup B^{\mathrm{c}}) \\ &\subseteq \mathrm{pCl}(A) \cap \mathrm{pCl}(B) \cap [\mathrm{pCl}(A^{\mathrm{c}}) \cup \mathrm{pCl}(B^{\mathrm{c}})] \\ &= [\mathrm{Fr}(A) \cap \mathrm{pCl}(B)] \cup [\mathrm{Fr}(B) \cap \mathrm{pCl}(A)] \\ &\subseteq \mathrm{Fr}(A) \cup \mathrm{Fr}(B).\end{aligned}$$

These derivations use distributivity and De Morgan's laws, without requiring Boolean complement identities.

6 Discussion

The proposed model was best viewed as a structural extension rather than a competing ranking algorithm. Relative to Fermatean fuzzy soft aggregation [9], soft expert models [10], complex N-soft structures [11], and bipolar soft models [12], the FTVNSS keeps two Fermatean truth–falsity pairs together with an explicit hesitation coordinate at each object-parameter pair. This preserves parameter-level uncertainty before any later scoring or aggregation step.

The comparison with recent Fermatean neutrosophic soft work was more direct. The models [20, 21] combined Fermatean and neutrosophic ideas within a soft-set environment, while the FDVNSS framework [22] additionally developed a soft topology. The present construction continued this topological line but imposed pairwise Fermatean admissibility and used a lattice order under which the null/absolute elements, complement, union, and intersection were mutually consistent. Hesitation remains an independent coordinate and is reversed by complementation. The explicit standing assumption before Theorem 5.11 also distinguishes ordinary pre-open behavior from the stronger finite-intersection property required by the later equalities.

Recent applied studies indicate several routes for turning the structure into a decision tool. Similarity and entropy measures [15] can guide future distance and information measures. Ranking designs [16, 19], healthcare selection methods [17], and Fermatean fuzzy soft aggregation for energy planning [18] provide additional templates for score and aggregation functions. The current results do not claim empirical superiority over those methods. Instead, the contribution lies in supplying a richer parameterized state space on which such procedures could be developed and evaluated.

For a future decision procedure, a scalar score should be tested for the distinctions it removes from the five-coordinate record, and an aggregation rule should make the roles of its parameters and weights explicit. These requirements concern extensions beyond the present topological theory. They will require dedicated score and aggregation definitions, followed by comparison and sensitivity studies.

The paper establishes algebraic and topological foundations but does not include a calibrated score function, sensitivity analysis, or benchmark dataset. The finite-intersection assumption on p-open sets is required for the specific identities identified above. Future work should characterize FTVNSTSs in which this condition follows from simpler axioms and test which weaker hypotheses suffice for individual results.

7 Conclusions and Future Directions

This paper formulated a FTVNSS with five explicit coordinates for truth, relative truth, hesitation, relative falsity, and falsity. The model preserved soft parameterization, imposed Fermatean cubic restrictions on the two truth–falsity pairs, and kept hesitation independent. The induced coordinate order yielded admissible complement, union, intersection, and AND/OR products, with the principal algebraic properties verified componentwise.

The set framework is extended to an FTVNSS topological space. Ordinary interior and closure, generalized open sets, p-interior, p-closure, exterior, frontier, and p-closed mappings are defined, and the assumptions required for finite-intersection and finite-union identities are made explicit. The distinctive feature is the joint use of soft parameters, a separate hesitation coordinate, and a topological layer.

Future work should develop score and accuracy functions for five-coordinate records, aggregation operators compatible with the set operations, and decision algorithms for multi-expert settings. Scalar projections should be checked for information loss, and parameter contributions should remain traceable during aggregation. These are design objectives for future procedures, rather than results established here. Empirical studies can then evaluate such procedures through benchmarks and sensitivity analysis. Dynamic and time-indexed FTVNSSs are further directions for domains with incomplete or conflicting evidence.

Author Contributions

Conceptualization, A.M.K.; methodology, F.K., A.M.K., and T.F.; validation, T.F.; formal analysis, F.K. and T.F.; investigation, F.K.; writing—original draft preparation, F.K.; writing—review and editing, A.M.K. and T.F.; supervision, A.M.K.; project administration, A.M.K. All authors have read and agreed to the published version of the manuscript.

Data Availability

No empirical dataset was generated or analyzed; all illustrative values used in the examples were reported in the manuscript.

Conflicts of Interest

The authors declare no conflicts of interest.

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